

TABLE OF CONTENTS

table of contents.....	1
1- Tools Required	2
2- Initial Conditions.....	2
3- Body TR Checks.....	2
4- Quad Head TR Checks	4
5- System Restoration	7
revision history.....	8

Description - This procedure ensures that the PIN Switch Driver Board and PIN diodes in the TR switch assemblies are not inadvertently damaged by transmitting via improper cable connections. It also isolates faulty PIN diodes (open or shorted diodes only; this procedure may not detect leaky diodes).

1- TOOLS REQUIRED

100-mHz Scope, Tektronix 468 or equivalent w/probe, 46-183029P61 or P64

2- INITIAL CONDITIONS



Equipment damage possibility. Do not excite the RF amplifier with no load connected to its output (i.e., ensure that RF amplifier input is disconnected from the digital attenuator output at CERD panel before initiating prescan). Permanent damage to amplifier components may otherwise occur.

3- BODY TR CHECKS

1. At ISE CERD (MR2A30A13), move switch to RF Out Disable to remove excitation from RF amplifier.
2. Disconnect the Body TR Driver output, J408, from the PIN Switch Driver Board.
3. Disable PIN Switch Driver Board Fault Detection by setting switch SW2 on the RF System Controller (RFSC) module to On (faults disabled).
4. Prepare system for body scan using protocol #11.1 shown in the Service Protocol list under Signa Help, or for alternate proprietary procedure, use Table 1. Landmark location is unimportant. No phantom is required for this test.

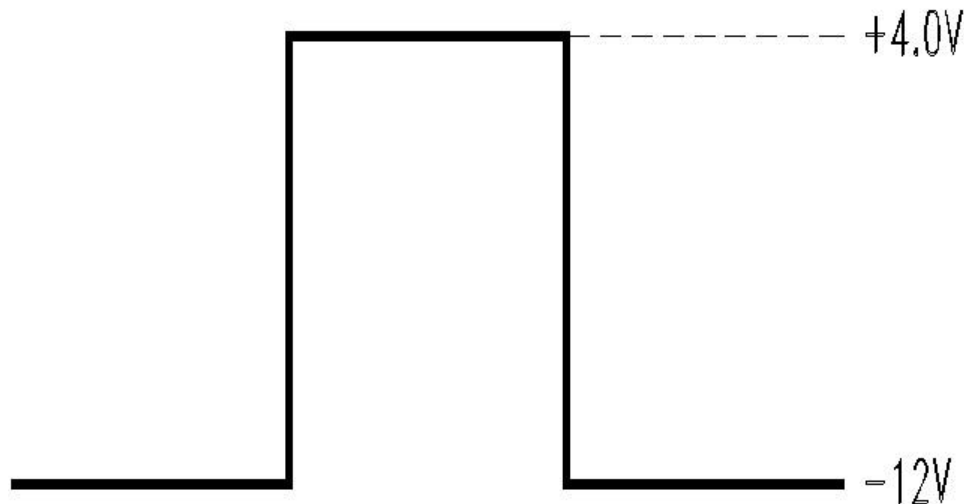
TABLE 1
BODY T/R CHECK PROTOCOL

Note:

This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

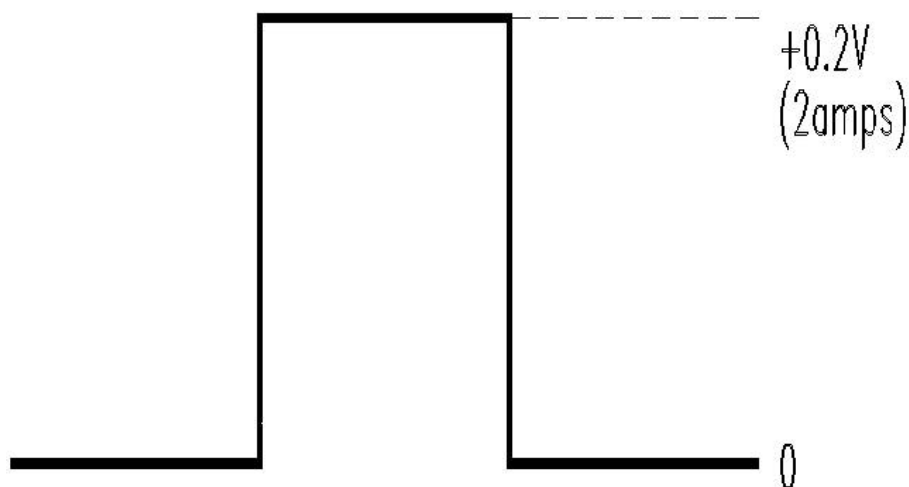
1. Click on **[New Pt]**, and enter
Id: **geservice**
Name: **t/r check**
Weight (Lb): **111**
Set Patient Protocols to **Service**.
2. **8.1:** In the protocol field, type **o.11.1** (o=Other, 1=body series) to load the protocol.
3. **[Save Series]**, then **[Prepare to Scan]**.

4. Select **[Manual Prescan]**, **[Scan TR]**.
5. Connect oscilloscope probe to TP16 on the PIN Switch Driver Board and verify waveform in Illustration L3171a. If you can not obtain this, check cables from PIN Switch Driver Board to system cabinet and verify the presence of unblank signal by checking TP43 on the PIN Switch Driver Board.



UNLOADED BODY TR DRIVER OUTPUT VOLTAGE WAVEFORM
ILLUSTRATION L3171A

6. Connect oscilloscope probe across R57 on the PIN Switch Driver Board.
7. Reconnect body TR driver output, J408, to the PIN Switch Driver Board and verify waveform on oscilloscope as shown in Illustration L3171b. If correct, continue with section 5-System Restoration.



QUAD II LOADED BODY TR DRIVER OUTPUT CURRENT WAVEFORM
ILLUSTRATION L3171B

8. If the magnitude of the positive voltage is less than shown in Illustration L3171b above, check for an open circuit:

- a. Check for open connection in cables from Body Quad Directional Coupler (MR1A10A1) to the Body Quadrature Hybrid assembly (MG3A3) located in rear pedestal.
- b. Check for open circuit in connectors of Quadrature Hybrid between body coil TR switch (MG3A3A2) and body attenuator TR switch (MG3A3A3). Then check cables between Attenuator TR Switch, Preamplifier TR Switch (MG3A3A4), and Preamplifier (MG3A3A5).

Note

Troubleshooting a body TR assembly - Quad II systems do not have series diodes, or individual diode boxes. The troubleshooting of a suspected bad body TR assembly consists of disassembly and checking (with an ohm meter) the individual diodes.

- c. If steps a and b are satisfactory, PIN diode in body coil TR switch (MG3A3A2) is probably open.
9. If the magnitude of the negative voltage is greater than shown in Illustration L3171b above, a short circuit may exist.
- a. Check cable terminations for short circuits between pin and shield.
 - b. If step a is satisfactory, check body TR switch diodes, bias resistors and capacitors for short circuit failures.
10. Select **[CANCEL]** to stop looping.

4- QUAD HEAD TR CHECKS

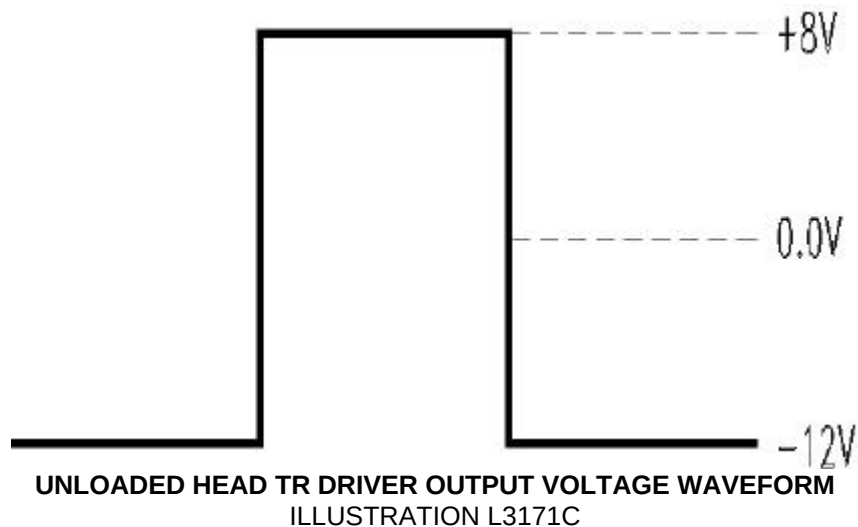
1. At ISE CERD (MR2A30A13) move switch to RF Out Disable to remove excitation from RF amplifier.
2. Disconnect the head TR driver output, J409, from the PIN Switch Driver Board.
3. Disable PIN Switch Driver Board Fault Detection by setting switch SW2 on PIN Switch Driver Board to On (faults disabled).
4. Prepare system for head scan using protocol #11.2 shown in the Service Protocol list under Signa Help, or for alternate proprietary procedure, per Table 2. Landmark location is unimportant. No phantom is required for this test.

TABLE 2
HEAD T/R CHECK PROTOCOL

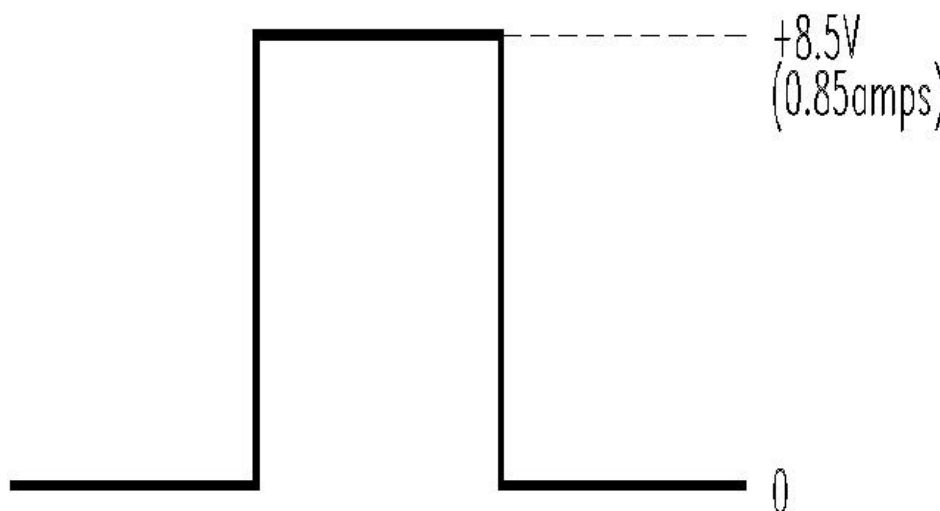
Note

This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

1. Click on **[New Pt]**, and enter
Id: **geservice**
Name: **t/r check**
Weight (Lb): **111**
Set Patient Protocols to **Service**.
2. In the protocol field, type **o.11.2** (o=Other, 2=head series) to load the protocol.
3. **[Save Series]**, then **[Prepare to Scan]**.
4. If you need help, see the User Interface Tutorial.
5. **[Manual Prescan]**, **[Scan TR]**.
6. Connect oscilloscope probe to TP9 on the PIN Switch Driver Board and verify waveform in Illustration L3171c. If you can not obtain this, check cables from PIN Switch Driver Board to system cabinet and verify the presence of unblank signal by checking TP43 on the PIN Switch Driver Board.



7. Connect oscilloscope probe to TP1 on the PIN Switch Driver Board.
8. Reconnect Head TR Driver output, J409, to the PIN Switch Driver Board and verify waveform on oscilloscope as shown in Illustration L3171d. If correct, continue with section 5- System Restoration.



LOADED HEAD TR DRIVER OUTPUT CURRENT WAVEFORM
ILLUSTRATION L3171D

9. If the magnitude of the positive voltage is less than shown in Illustration L3171c above, check for an open circuit:
 - a. Check for open connection in cables from Head Quad Directional Coupler (MR1A10A1) to the head TR assembly (MG2A16A5), located in the Head Coil Carriage Assembly (MG2A16).
 - b. Check for open circuit in connectors on the head TR assembly (MG2A16A5). Then check cables between head TR assembly and Head Preamplifier module (MG2A16A3).
 - c. If steps a and b are satisfactory, PIN diode in head coil TR switch (MG2A16A5) is probably open.
10. If the magnitude of the negative voltage is greater than shown in Illustration L3171b in Section 3, check for short circuit:
 - a. Disconnect the head coil by disconnecting at the quick disconnect (MG2A16A4). If short circuit signal stops, check quick disconnect assembly and the jumpers on the Head Coil Isolation Network Board (MG2A16A1A1).
 - b. Check cable terminations for short circuits between pin and shield.
 - c. If step a is satisfactory, check head TR switch diodes, bias resistors and capacitors for short circuit failures.
11. Select **[CANCEL]** To stop looping.

5- SYSTEM RESTORATION

1. At ISE CERD (MR2A30A13), move switch to RF Out Normal.
2. Verify that all cables that were temporarily disconnected are restored to original configuration.
3. Enable PIN Switch Driver Board Fault Detection by setting switch SW2 on the PIN Switch Driver Board to Off (faults not disabled).

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 14, 1998	J. Saperstein	Initial conversion from Toolbook to Word.